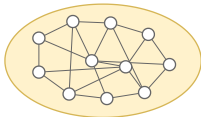


a**yeast in nature**

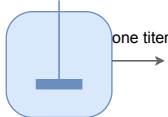
robust and cross-coupled;
fourteen times the
curated model's reaction
space, at weaker affinities

engineer

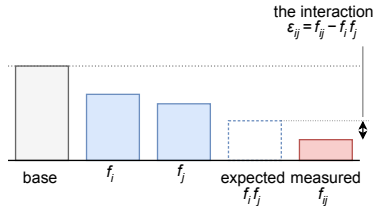
delete or overexpress a
few genes; keep the
strain that makes more of
one product



the coupling that made
the cell robust is still
there, and is not
measured

yeast in a bioreactor

one product, one number
per strain; what pays off
in nature is not what a
factory needs

b

On yeast growth, genome-wide:
~550,000 negative to ~350,000
positive digenic interactions, and
~100 times as many trigenic,
~25% weaker and scored
negative.

Predicted, an interaction is a step
a design can climb. Unpredicted,
a campaign is left to trial and
error, and the combinations that
raise a product are the rarer sign.